Enhancing License Plate Detection and Character Recognition Using YOLO, CNN, and Tesseract OCR

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Git hub : https://github.com/pp23aai/MSc-Final-project--2024-25

Project Title :" **Enhancing License Plate Detection and Character Recognition Using YOLO, CNN, and Tesseract OCR"**

- **A Research question?**
  What processing techniques can effectively enhance the accuracy and speed of automatic license plate detection and character recognition using YOLO, CNN, and Tesseract OCR?

- **Summary of the project**
  Automatic license plate detection and character recognition are essential technologies used in areas like traffic enforcement, parking management, and security. As the demand for more accurate and efficient systems grows, this project aims to explore methods to improve their performance.

- **Project Focus:** This study will evaluate three key techniques:

- **YOLO (You Only Look Once):** A fast and precise object detection algorithm. YOLO will be used to quickly locate and identify license plates in real time ssess its accuracy and processing speed.

- **CNN (Convolutional Neural Networks):** A type of neural network known for its strength in image recognition. In this project, CNNs will be used to enhance the accuracy of character recognition on detected plates. Analyze CNN for Character Recognition

- **Tesseract OCR:** An open-source Optical Character Recognition tool. Tesseract will be used as a baseline to compare its performance in recognizing characters against CNN models.

- Compare Performance Metrics: Analyze and compare the accuracy, speed, and robustness of YOLO, CNN, and Tesseract OCR in the context of license plate detection and recognition.

- Provide Insights and Recommendations: Draw conclusions from the comparative analysis to suggest best practices for enhancing license plate detection and character recognition systems.

- **Reference List :**
  Slimani, I., Zaarane, A., Al Okaishi, W., Atouf, I., & Hamdoun, A. (Year). *An automated license plate detection and recognition system based on wavelet decomposition and CNN*. https://www.sciencedirect.com/science/article/pii/S2590005620300254

- **Chang, S. L., Chen, L. S., Chung, Y. C., & Chen, S. W. (2004).** *Automatic License Plate Recognition*. IEEE Transactions on Intelligent Transportation Systems, 5(1), 42-53. https://www.researchgate.net/publication/3427867_Automatic_License_Pla te_Recognition

- **Qadri, M. T., & Asif, M. (Year).** *Automatic Number Plate Recognition System for Vehicle Identification Using Optical Character Recognition*. IEEE. https://ieeexplore.ieee.org/document/5169511

Project Timeline

Data Collection and Preprocessing (Oct 18 – Nov 1):

- **Collect Data:** Download the license plate images from Robo flow.
- **Data Cleaning:** Handle missing or low-quality images. Make sure all images are resized, formatted correctly, and normalized to prepare for model training.
- **Visualize Data:** Perform exploratory data analysis (EDA) by checking sample images and analyzing data distribution patterns, like the number of characters on plates, image quality, etc.

YOLO Model Implementation (Nov 1 – Nov 15):

- **Train YOLO:** Set up YOLOv11 to detect license plates in images. Train it using the cleaned and prepared dataset.
- **Evaluate YOLO:** Measure performance using metrics like accuracy, precision, and recall. Fine-tune if necessary by adjusting learning rates or other parameters.

CNN and Tesseract OCR Implementation (Nov 15 – Nov 29):

- **Train CNN:** Use CNN to recognize and classify characters on detected license plates. Train the CNN model using labeled images of characters.
- **Use Tesseract:** Implement Tesseract OCR to recognize characters and compare its performance with CNN to see which model gives better results.

Model Comparison and Optimization (Nov 29 – Dec 6):

- **Compare YOLO and Character Recognition Models:** Evaluate how well YOLO detects plates and compare CNN vs. Tesseract for recognizing characters. Check accuracy, precision, recall, and speed.
- **Fine-tune Models:** Improve the performance of YOLO and CNN by adjusting parameters like learning rates or adding more layers to the models.

Result Analysis and Reporting (Dec 6 – Dec 20):

- **Analyze Results:** Summarize which model performed better and why. Highlight strengths and weaknesses for each model.
- **Document Findings:** Write down your analysis, including how each model worked and what could be improved.

Final Report and Presentation (Dec 20 – Jan 3):

- **Prepare the Report:** Write the final project report covering the methodology, results, and conclusions.
- **Create the Presentation:** Design presentation slides summarizing your project and key results for final submission.

Project Management Plan

Overview dataset:

- Dataset name : ALPR (Automatic License Plate Recognition) Dataset alpr - v1 2021-12-31 1:53am (roboflow.com)
- Purpose: This dataset contains annotated images of vehicle license plates captured under various conditions, aimed at training models for accurate license plate detection and character recognition

Details about the Dataset

- Time Period: The dataset includes a diverse set of images taken in different lighting conditions and angles, ensuring robustness for model training.
- Data Type: The dataset consists of image data (e.g., PNG, JPG) with annotations for bounding boxes (for plate detection) and corresponding text labels (for character recognition).
- Data Format: The data will be in image formats, along with corresponding annotation files.
- Data Size: The dataset consists of 9,000 images, providing a solid foundation for training and evaluating models.

Data Fields: Image File: Contains the vehicle license plate images.

- Bounding Boxes: Coordinates for the detected license plates in each image.
- Character Labels: Textual data indicating the characters on the license plates.

Document control

- Version Control: Code and project files will be managed using GitHub. ○ Weekly commits will be made to track changes and updates. GitHub Repository: [Insert your actual GitHub link here].
- ReadMe File The ReadMe file in the GitHub repository will include: A summary of the project and its objectives.
- Instructions on how to set up the environment, run the code, and replicate the results.
- Backup Frequency: Weekly backups of both the dataset and project files will be made to GitHub and OneDrive to ensure the safety of the data.
- Data Sharing: Data and code will be shared with supervisors and peers via GitHub, providing controlled access.○ OneDrive will be used for additional backup storage and ease of access

Ethical Requirements

1. GDPR Compliance: The dataset does not contain personal data, ensuring compliance with GDPR regulations.
2. University Ethical Policies: The project adheres to the University of Hertfordshire's ethical policies, ensuring responsible use of data.
3. Permission to Use Data: The dataset is publicly available through Robo flow, ensuring legal permissions for use in research.
4. Ethical Data Collection: The data is collected from reputable sources, ensuring that it has been gathered ethically and can be freely used for academic purposes